

## CODEVEDX AI/ML Internship Report

### Project 1: Utility Usage Prediction Tool (Machine Learning)

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#### Acknowledgement

I would like to express my sincere gratitude to CODEVEDX for providing me with the opportunity to complete this Artificial Intelligence & Machine Learning internship. This project has helped me enhance my knowledge of Python programming, data handling, Machine Learning concepts, and software development practices.

#### Abstract

The **Utility Usage Prediction Tool** is a Machine Learning-based console application developed using Python. The application allows users to manage electricity usage records by adding, updating, deleting, and viewing data stored in a CSV file.

The project uses the **Linear Regression** algorithm to analyze historical electricity consumption and predict future electricity usage. It also demonstrates important programming concepts such as file handling, exception handling, data manipulation using Pandas, and Machine Learning model implementation using Scikit-learn.

This project provides a simple and effective solution for predicting utility consumption while improving practical knowledge of Artificial Intelligence and Machine Learning.

#### Introduction

Electricity consumption is continuously increasing due to the growing use of electrical appliances in homes, industries, and commercial buildings. Predicting future electricity usage helps organizations plan energy production, reduce wastage, and improve resource management.

The **Utility Usage Prediction Tool** is designed to store daily electricity consumption data and estimate future electricity usage based on previous records. The project combines Python programming with Machine Learning techniques to create a user-friendly console application.

It demonstrates how historical data can be used to build predictive models for real-world applications.

## Objectives

The main objectives of this project are:

- To develop a menu-driven console application using Python.
- To store electricity usage records in a CSV file.
- To implement CRUD (Create, Read, Update, Delete) operations.
- To apply the Linear Regression Machine Learning algorithm.
- To predict future electricity usage using historical data.
- To understand data preprocessing and model training.
- To improve programming and problem-solving skills.
- To gain practical knowledge of Artificial Intelligence and Machine Learning.

## Problem Statement

Manual analysis of electricity consumption records is time-consuming and inefficient. Predicting future electricity usage manually becomes difficult as the amount of data increases.

The objective of this project is to automate the management of electricity usage records and use Machine Learning techniques to accurately predict future electricity consumption. This helps users make better decisions regarding energy planning and consumption.

## Software Requirements

The following software was used to develop this project:

- Operating System: Windows 10 / Windows 11
- Programming Language: Python 3.x
- IDE: Visual Studio Code (VS Code)
- Libraries:
  - Pandas
  - Scikit-learn
  - OS
  - Module

- CSV File for Data Storage
- Command Prompt / VS Code Terminal

### Hardware Requirements

The minimum hardware requirements for running this project are:

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: 500 MB free storage
- Keyboard and Mouse
- Monitor
- Internet Connection (for installing Python libraries)

### Technologies Used

The project was developed using the following technologies:

| Technology         | Purpose                                    |
|--------------------|--------------------------------------------|
| Python             | Main programming language                  |
| Pandas             | Reading, writing and manipulating CSV data |
| Scikit-learn       | Building the Linear Regression model       |
| CSV File           | Storing electricity usage records          |
| Linear Regression  | Predicting future electricity usage        |
| VS Code            | Writing and executing Python code          |
| Exception Handling | Handling runtime errors gracefully         |

## Algorithm

### Step 1

Start the application.

### Step 2

Load the electricity usage data from **Project1.csv**.

### Step 3

Display the main menu to the user.

### Step 4

Allow the user to perform operations such as:

- Add Data
- Update Data
- Delete Data
- View Data

### Step 5

Train the Linear Regression model using the available dataset.

### Step 6

Accept the future day as input from the user.

### Step 7

Predict the electricity usage for the entered day.

### Step 8

Display the predicted electricity usage.

### Step 9

Repeat the menu until the user selects Exit.

### Step 10

Terminate the application.

## Dataset

File: Project1.csv

Columns: Day, Electricity\_Usage

## Output Screenshots

- Main menu

```
UTILITY USAGE PREDICTION TOOL
1. Add Data
2. Update Data
3. Delete Data
4. View Data
5. Predict Usage
6. Exit
Enter Choice:
```

- Add Data

```
UTILITY USAGE PREDICTION TOOL
1. Add Data
2. Update Data
3. Delete Data
4. View Data
5. Predict Usage
6. Exit
Enter Choice: 1
Enter Day: 31
Enter Electricity Usage: 205
Data Added Successfully.
```

- View Data

```
Enter Choice: 4
Day Electricity_Usage
0 1 120.0
1 2 125.0
2 3 130.0
3 4 128.0
4 5 135.0
5 6 138.0
6 7 142.0
7 8 145.0
8 9 148.0
9 10 150.0
10 11 155.0
11 12 158.0
12 13 160.0
13 14 162.0
14 15 165.0
15 16 168.0
16 17 170.0
17 18 172.0
18 19 175.0
19 20 178.0
20 21 180.0
21 22 183.0
22 23 185.0
23 24 188.0
24 25 190.0
25 26 193.0
26 27 195.0
27 28 198.0
28 29 200.0
29 30 203.0
30 31 205.0
```

- Update Data

```
Enter Choice: 2
  Day Electricity_Usage
0    1          120.0
1    2          125.0
2    3          130.0
3    4          128.0
4    5          135.0
5    6          138.0
6    7          142.0
7    8          145.0
8    9          148.0
9   10          150.0
10  11          155.0
11  12          158.0
12  13          160.0
13  14          162.0
14  15          165.0
15  16          168.0
16  17          170.0
17  18          172.0
18  19          175.0
19  20          178.0
20  21          180.0
21  22          183.0
22  23          185.0
23  24          188.0
24  25          190.0
25  26          193.0
26  27          195.0
27  28          198.0
28  29          200.0
29  30          203.0
30  31          205.0
Enter Day to Update: 31
Enter New Usage: 208
Data Updated Successfully.
```

- Delete Data

```
Enter Choice: 3
  Day Electricity_Usage
0    1          120.0
1    2          125.0
2    3          130.0
3    4          128.0
4    5          135.0
5    6          138.0
6    7          142.0
7    8          145.0
8    9          148.0
9   10          150.0
10  11          155.0
11  12          158.0
12  13          160.0
13  14          162.0
14  15          165.0
15  16          168.0
16  17          170.0
17  18          172.0
18  19          175.0
19  20          178.0
20  21          180.0
21  22          183.0
22  23          185.0
23  24          188.0
24  25          190.0
25  26          193.0
26  27          195.0
27  28          198.0
28  29          200.0
29  30          203.0
30  31          208.0
Enter Day to Delete: 31
Data Deleted Successfully.
```

- Prediction Output

```
UTILITY USAGE PREDICTION TOOL
1. Add Data
2. Update Data
3. Delete Data
4. View Data
5. Predict Usage
6. Exit
Enter Choice: 5
Enter Future Day: 32
Predicted Electricity Usage: 210.37 Units
```

## Conclusion

The **Utility Usage Prediction Tool** was successfully developed using Python and the **Linear Regression** algorithm. The application allows users to manage electricity usage records and predict future electricity consumption based on historical data.

This project enhanced my understanding of Python programming, CSV file handling, data analysis, and basic Machine Learning concepts.

## References

- [Python Official Documentation](#)
- [Pandas Official Documentation](#)
- [Scikit-learn Official Documentation](#)
- [NumPy Official Documentation](#)
- [Visual Studio Code Documentation](#)